

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Indicative content 1. Oxalic acid has two acidic protons 2. The K_a /acidity/ease of removal of each proton is different so they give two different vertical regions 3. First vertical region occurs at half the volume of second as each occurs after removing same number of protons 4. Flat regions from 5-10 and 20-25 due to formation of buffer 5. Alice, Brychan and David methods will work 6. Indicators only work if they change colour within a vertical region 7. Alice's method allows the first and second equivalence points to be found by plotting a curve 8. Brychan's indicator will change colour fully during the second vertical region 9. David's indicator will change colour fully during the first vertical region 10. Carys's method will not work as indicator will change colour gradually as sodium hydroxide is added						
						3	3	6		3